1) Integration

a) Integration

If $\frac{dy}{dx} = x^n$, then $y = \int x^n \, dx = \frac{x^{n+1}}{n+1} + c \ (n \neq -1)$

If $\frac{dy}{dx} = ax^n$, then $y = \int ax^n \, dx = \frac{ax^{n+1}}{n+1} + c \ (n \neq -1)$

Example 1

Given $\frac{dy}{dx} = 9x^2 - 4x + 3x^{\frac{1}{2}} - 7$, find y.

$y = \int (9x^2 - 4x + 3x^{\frac{1}{2}} - 7)\,dx$ — We need to integrate $\frac{dy}{dx}$ to get y.

$= \frac{9x^3}{3} - \frac{4x^2}{2} + \frac{3x^{\frac{3}{2}}}{\frac{3}{2}} - 7x + c$ — Use $\int x^n \, dx = \frac{x^{n+1}}{n+1}$ for each term.

$= 3x^3 - 2x^2 + 2x^{\frac{3}{2}} - 7x + c$ — Simplify where possible and include '+ c' in your answer.

b) Finding constant of integration

Example 3

Find the equation of the curve that passes through the point $(4, -1)$ and where $\frac{dy}{dx} = \frac{5x^2+1}{\sqrt{x}}$.

$\frac{dy}{dx} = \frac{5x^2+1}{\sqrt{x}} = 5x^{\frac{3}{2}} + x^{-\frac{1}{2}}$ — Divide each term by $x^{\frac{1}{2}}$.

$y = \frac{5x^{\frac{5}{2}}}{\frac{5}{2}} + \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + c$ — Integrate each term and include '+ c' at the end.

$= 2x^{\frac{5}{2}} + 2x^{\frac{1}{2}} + c$ — Dividing by $\frac{5}{2}$ is the same as multiplying by $\frac{2}{5}$.

As $(4, -1)$ lies on the curve,

$-1 = 2(4)^{\frac{5}{2}} + 2(4)^{\frac{1}{2}} + c$ — Substitute $x = 4$, $y = -1$.

$c = -1 - 64 - 4 = -69$ — Find the value of c.

$y = 2x^{\frac{5}{2}} + 2x^{\frac{1}{2}} - 69$ — Write down the equation of the curve.

Note: You can check you have integrated correctly by differentiating,

e.g. $\frac{d}{dx}\left(2x^{\frac{5}{2}} + 2x^{\frac{1}{2}} + c\right) = 5x^{\frac{3}{2}} + x^{-\frac{1}{2}}$

Example 4

A curve is such that $f'(x) = \frac{10}{x^3} - 4$ and the point $(-1, 2)$ lies on the curve.
Find the equation of the curve.

$f(x) = \int\left(\frac{10}{x^3} - 4\right)dx = \int(10x^{-3} - 4)\,dx$ — Integrate $f'(x)$ to find $f(x)$ by expressing each term as a power of x.

$= \frac{10x^{-2}}{-2} - 4x + c$ — Integrate term by term.

$= -\frac{5}{x^2} - 4x + c$ — Simplify and write your negative powers as fractions.

$2 = -\frac{5}{(-1)^2} - 4(-1) + c$ — Substitute for $x = -1$ and $y = 2$.

$c = 3$ — Find the value of c.

$f(x) = -\frac{5}{x^2} - 4x + 3$ — Write down the equation of the curve.

Example 5

A curve is such that $\frac{d^2y}{dx^2} = -6x$ and the curve has a maximum point at $(1, 2)$.
Find the equation of the curve.

$\frac{dy}{dx} = \int -6x\,dx$ — Integrate $\frac{d^2y}{dx^2}$ to find $\frac{dy}{dx}$.

$= -3x^2 + c$

$0 = -3(1)^2 + c$ — When $x = 1$ and $y = 2$ we have a maximum point, and so $\frac{dy}{dx} = 0$.

$c = 3$

$\frac{dy}{dx} = -3x^2 + 3$ — We now have an equation for $\frac{dy}{dx}$.

$y = -x^3 + 3x + k$ — Integrate to find y.

$2 = -(1)^3 + 3(1) + k$ — $y = 2$ when $x = 1$.

$k = 0$

$y = -x^3 + 3x$ — We can now write an equation for the curve.

Example 6

Find $\int (4x-3)^5 \, dx$.

$$\int (4x-3)^5 \, dx = \frac{1}{4}\frac{(4x-3)^{5+1}}{(5+1)} + c$$

Use the standard result with $a = 4$, $b = -3$, $n = 5$.

$$= \frac{(4x-3)^6}{24} + c$$

Example 7

Find $\int \frac{4}{(1-2x)^7} \, dx$.

$$\int \frac{4}{(1-2x)^7} \, dx = 4\int (1-2x)^{-7} \, dx$$

We can take the constant factor 4 outside the integral.

$$= 4 \times \frac{1}{-2} \times \frac{(1-2x)^{-6}}{-6} + c$$

$a = 1$, $b = -2$, $n = -7$

$$= \frac{(1-2x)^{-6}}{3} + c$$

$$= \frac{1}{3(1-2x)^6} + c$$

Example 8

$\int \frac{6}{\sqrt{(4x+9)}} \, dx$

$$\int \frac{6}{\sqrt{(4x+9)}} \, dx = \int 6(4x+9)^{-\frac{1}{2}} \, dx$$

Write in a form using negative powers.

$$= 6 \times \frac{1}{4}\frac{(4x+9)^{\frac{1}{2}}}{\frac{1}{2}} + c$$

Use the standard result with $a = 4$, $b = 9$, $n = -\frac{1}{2}$.

$$= 3\sqrt{(4x+9)} + c$$

Simplify your answer.

c) Definite integrals

Example 9

a) Find $\int (3x-2)^8 \, dx$. **b)** Hence find $\int_0^1 (3x-2)^8 \, dx$.

a) $\int (3x-2)^8 \, dx = \frac{1}{3}\frac{(3x-2)^9}{9} + c$ ← Also divide by the coefficient of x.

$= \frac{(3x-2)^9}{27} + c$

b) $\int_0^1 (3x-2)^8 \, dx = \left[\frac{(3x-2)^9}{27}\right]_0^1$ ← We use square brackets to show we have integrated and we write the limits in the positions shown.

$= \frac{(3(1)-2)^9}{27} - \frac{(3(0)-2)^9}{27}$ ← Calculate $f(b) - f(a)$.

$= \frac{1}{27} + \frac{512}{27} = \frac{513}{27} = 19$

Example 10

Find $\int_1^4 \frac{x^2 + x^3}{\sqrt{x}} \, dx$.

$\int_1^4 \frac{x^2 + x^3}{\sqrt{x}} \, dx = \int_1^4 \left(x^{\frac{3}{2}} + x^{\frac{5}{2}}\right) dx$ ← Divide each term in the numerator by $x^{\frac{1}{2}}$.

$= \left[\frac{2}{5}x^{\frac{5}{2}} + \frac{2}{7}x^{\frac{7}{2}}\right]_1^4$

$= \left(\frac{2}{5}4^{\frac{5}{2}} + \frac{2}{7}4^{\frac{7}{2}}\right) - \left(\frac{2}{5}1^{\frac{5}{2}} + \frac{2}{7}1^{\frac{7}{2}}\right)$ ← Substitute $x = 4$ and $x = 1$.

$= \left(\frac{2}{5} \times 32 + \frac{2}{7} \times 128\right) - \left(\frac{2}{5} + \frac{2}{7}\right)$ ← Evaluate as separate brackets.

$= 12\frac{2}{5} + 36\frac{2}{7} = 48\frac{24}{35}$

Exercise 10.4

1. Find

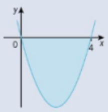
a) $\int_0^1 6x \, dx$ **b)** $\int_2^5 x^2 \, dx$ **c)** $\int_4^9 \sqrt{x} \, dx$

d) $\int_{-1}^1 \frac{x}{2} \, dx$ **e)** $\int_1^2 \frac{3}{x^2} \, dx$ **f)** $\int_1^4 \frac{1}{\sqrt{x}} \, dx$.

d) Area bounded between two curves

Example 12

Find the area bounded by the curve with equation $y = x(x - 4)$ and the x-axis.



First draw a sketch.

$$\text{Area} = \int_0^4 (x^2 - 4x)\,dx$$

$$= \left[\frac{1}{3}x^3 - 2x^2\right]_0^4$$

$$= \left(\frac{64}{3} - 32\right) - (0 - 0) = -10\frac{2}{3}$$

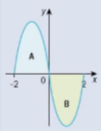
Note: Where the value of the integral is negative this shows that the region is below the x-axis.

$$\text{Area} = 10\frac{2}{3}$$

Give the area as a positive value.

Example 13

Find the area bounded by the curve with equation $y = x(x + 2)(x - 2)$ and the x-axis.



First draw a sketch.

We find the area by considering the two separate regions A and B.

$$\text{Area A} = \int_{-2}^0 (x^3 - 4x)\,dx$$

Expand the brackets.

$$= \left[\frac{1}{4}x^4 - 2x^2\right]_{-2}^0$$

$$= (0 - 0) - (4 - 8) = 4$$

▶ Continued on the next page

$$\text{Area B} = \int_0^2 (x^3 - 4x)\,dx$$

We can use the result from region A above.

$$= \left[\frac{1}{4}x^4 - 2x^2\right]_0^2$$

$$= (4 - 8) - (0 - 0) = -4$$

The negative value shows the area is under the x-axis.

Total area $= 4 + 4 = 8$

Note that if we calculate $\int_{-2}^2 (x^3 - 4x)\,dx$ we get 0 as the answer.

We can also use this technique to find the area bounded by a curve, $x = f(y)$, the y-axis and the lines $y = a$ and $y = b$.

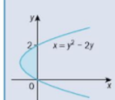
The area between the curve $x = f(y)$, the **y-axis**, and the lines $y = a$ and $y = b$ is given by:

$$\text{Area} = \int_{y=a}^{y=b} x\,dy \text{ or } \int_{y=a}^{y=b} f(y)\,dy$$

Note: We usually write $\int_a^b$ instead of $\int_{y=a}^{y=b}$

Example 14

Find the area bounded by the curve with equation $x = y^2 - 2y$ and the y-axis as shown in the diagram.



$$\text{Area} = \int_{y=a}^{y=b} x\,dy$$

Use the correct formula.

$$= \int_0^2 (y^2 - 2y)\,dy$$

Everything is using y, including the limits.

$$= \left[\frac{1}{3}y^3 - y^2\right]_0^2$$

$$= \left(\frac{8}{3} - 4\right) - (0 - 0) = -\frac{4}{3}$$

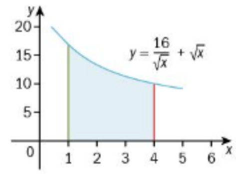
The negative value shows that the required area is **to the left** of the y-axis.

We write that the required area is $\frac{4}{3}$.

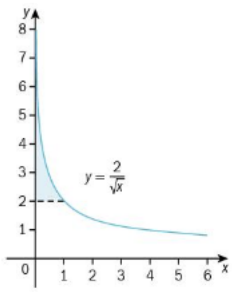
Write as a positive value.

8. The diagram shows part of the curve with equation $y = \dfrac{16}{\sqrt{x}} + \sqrt{x}$.

Find the area bounded by the curve, the lines $x = 1$, $x = 4$ and the x-axis.



9.



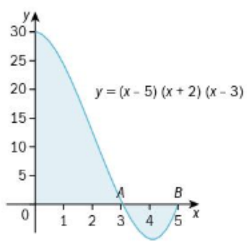
Find the area of the shaded region.

10. The diagram shows part of the curve with equation $y = (x - 5)(x + 2)(x - 3)$.

a) Write down the coordinates of A and B.

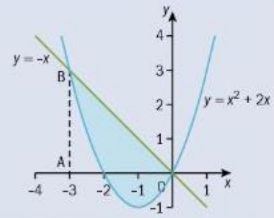
b) Show that the equation may be written as $y = x^3 - 6x^2 - x + 30$.

c) Find the total shaded area.



Example 16

The diagram shows the curve $y = x^2 + 2x$ and the line $y = -x$.
Work out the area of the shaded region.



The curve meets the line when $x^2 + 2x = -x$

$x^2 + 3x = 0$

$x(x + 3) = 0$

Solve the two equations simultaneously to find where the line intersects the curve.

They meet at $(0, 0)$ and $(-3, 3)$

Area of triangle 0AB $= \frac{1}{2}(3 \times 3) = \frac{9}{2}$

Area under the curve, between $x = -3$ and $x = -2$

$$= \int_{-3}^{-2} (x^2 + 2x)\,dx$$

The dotted line is at $x = -3$.

$$= \left[\frac{1}{3}x^3 + x^2\right]_{-3}^{-2}$$

$$= \left(-\frac{8}{3} + 4\right) - (-9 + 9)$$

$$= \frac{4}{3}$$

Area of shaded region above the x-axis $= \frac{9}{2} - \frac{4}{3} = \frac{19}{6}$

Area of shaded region below the x-axis

$$= \int_{-2}^{0} (x^2 + 2x)\,dx$$

This is the area between the curve and the x-axis.

$$= \left[\frac{1}{3}x^3 + x^2\right]_{-2}^{0}$$

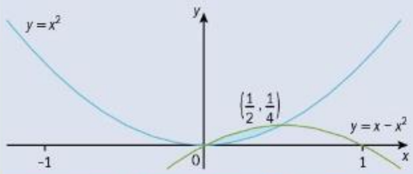
$$= (0 + 0) - \left(-\frac{8}{3} + 4\right) = -\frac{4}{3}$$

The area is a negative value as it is below the curve.

Area required $= \frac{19}{6} + \frac{4}{3} = \frac{9}{2}$

Example 17

Find the area of the finite region bounded by $y = x^2$ and $y = x - x^2$.



$$\text{Area} = \int_0^{\frac{1}{2}} (x - x^2)\,dx - \int_0^{\frac{1}{2}} x^2\,dx$$

Area under $y = x - x^2$ minus area under $y = x^2$

$$= \int_0^{\frac{1}{2}} (x - x^2 - x^2)\,dx$$

Combine the two integrals.

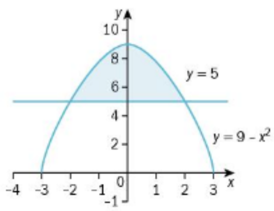
$$= \int_0^{\frac{1}{2}} (x - 2x^2)\,dx$$

$$= \left[\frac{1}{2}x^2 - \frac{2}{3}x^3\right]_0^{\frac{1}{2}}$$

$$= \left(\frac{1}{8} - \frac{1}{12}\right) - (0 - 0)$$

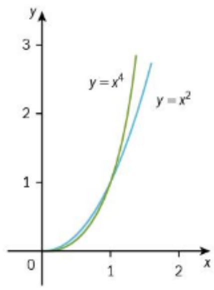
$$= \frac{1}{24}$$

3.



The diagram shows the curve $y = 9 - x^2$ and the line $y = 5$.
Find the shaded area.

4.



The diagram shows parts of the curves $y = x^2$ and $y = x^4$.
Find the shaded area.

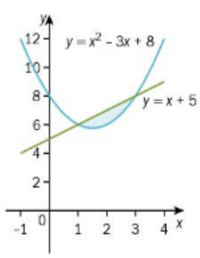
5. Find the area between the curve with equation $y = x(x - 1)$ and the line $y = 3x$.

Find the coordinates of the points of intersection and sketch the graphs.

6. **a)** Show that the curve $y = 2x^2 + 3$ and the curve $y = 10x - x^2$ meet at the points where $x = 3$ and $x = \frac{1}{3}$.

b) Find the area between these two curves.

7.



Find the area bounded by the curve $y = x^2 - 3x + 8$ and the line $y = x + 5$.

e) Improper integrals

x

For **(1)** **i)**

We define the **improper integral** $\int_a^\infty f(x)dx$ as $\lim_{b\to\infty}\int_a^b f(x)dx$, provided the limit exists.

ii)

We define the **improper integral** $\int_{-\infty}^b f(x)dx$ as $\lim_{a\to-\infty}\int_a^b f(x)dx$, provided the limit exists.

For **(2)**

When $f(x)$ is defined for $0 < x < b$, but $f(x)$ is not defined when $x = 0$, then the **improper integral** $\int_0^b f(x)dx = \lim_{a\to 0^+}\int_a^b f(x)dx$, provided the limit exists.

Note: $\lim_{a\to 0^+}$ denotes 'a tends to 0 from just above 0'.

Example 18

Show that the improper integral $\int_1^\infty \frac{1}{x^2\sqrt{x}}dx$ has a value and find that value.

$$\int_1^b \frac{1}{x^2\sqrt{x}}dx = \int_1^b x^{-\frac{5}{2}}dx$$

Write the integral with an upper limit, say b (where b is a large number).

$$= \left[-\frac{2}{3x^{\frac{3}{2}}}\right]_1^b$$

Integrate the function in the normal way.

$$= \left(-\frac{2}{3b^{\frac{3}{2}}}\right) - \left(-\frac{2}{3}\right)$$

As $b \to \infty$, $\frac{2}{3b^{\frac{3}{2}}} \to 0$

As b tends to ∞, the denominator gets larger and the value of the fraction tends to 0.

Thus $\int_1^\infty \frac{1}{x^2\sqrt{x}}dx = -0 + \frac{2}{3}$

Hence the improper integral does exist (as it has a finite value) and has a value of $\frac{2}{3}$.

Example 19

Show that only one of the following improper integrals has a finite value and find that value.

a) $\int_{-2}^0 \frac{2}{x^5}dx$ **b)** $\int_{-\infty}^{-2} \frac{2}{x^5}dx$

a) $$\int_{-2}^p \frac{2}{x^5}dx = \left[-\frac{2}{4x^4}\right]_{-2}^p$$

As the function to be integrated is not defined at $x = 0$, replace the upper limit 0 with a letter, say p ($-2 < p < 0$ and p just less than 0).

$$= \left(-\frac{1}{2p^4}\right) - \left(-\frac{1}{2(-2)^4}\right)$$

As $p \to 0$, $\frac{1}{2p^4}$ has no finite value

As $p \to 0$, $\frac{1}{2p^4} \to \infty$ and so has no finite value.

and so, $-\frac{1}{2p^4} + \frac{1}{32}$ has no finite value.

We can say the integral does not exist.

b) $$\int_q^{-2} \frac{2}{x^5}dx = \left[-\frac{2}{4x^4}\right]_q^{-2}$$

Start by replacing $-\infty$ with a variable, say q.

$$= \left(-\frac{1}{2(-2)^4}\right) - \left(-\frac{1}{2q^4}\right)$$

As $q \to -\infty$, $\frac{1}{2q^4} \to 0$

Thus $\int_{-\infty}^{-2} \frac{2}{x^5}dx = -\frac{1}{32} + 0$

Hence, this improper integral does exist and has a value of $-\frac{1}{32}$.

3. a) Find, in terms of a and b, the value of the integral $\int_a^b \frac{6}{x^{\frac{5}{2}}}\,dx$.

b) Show that only one of the following improper integrals has a finite value and find that value.

i) $\int_0^9 \frac{6}{x^{\frac{5}{2}}}\,dx$ ii) $\int_9^\infty \frac{6}{x^{\frac{5}{2}}}\,dx$

4. For each of the following improper integrals, find the value of the integral or explain briefly why it does not have a value.

a) $\int_{27}^\infty \sqrt[3]{x}\,dx$ b) $\int_{27}^\infty \frac{1}{\sqrt[3]{x^4}}\,dx$

5. Explain briefly why $\int_0^{100} 3x^{-\frac{1}{2}}dx$ is an improper integral.

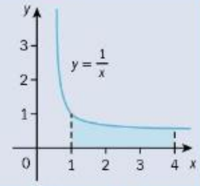
6. For each of the following improper integrals, find the value of the integral or explain briefly why it does not have a value.

a) $\int_0^{100} 3x^{-\frac{1}{2}}\,dx$ b) $\int_0^{100} 3x^{-\frac{3}{2}}\,dx$ c) $\int_{100}^\infty 3x^{-\frac{1}{2}}\,dx$

f) Volumes of revolution

Example 20

Find the volume obtained when the shaded region is rotated through 360° about the x-axis.



$$V = \int_1^4 \pi y^2 \, dx = \pi \int_1^4 \frac{1}{x^2} \, dx$$

$y^2 = \left(\frac{1}{x}\right)^2$ and $a = 1$, $b = 4$

$$= \pi \left[-\frac{1}{x} \right]_1^4$$

π is a constant so can be taken outside the integral.

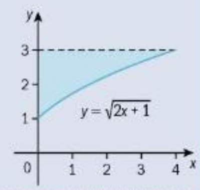
$$= \pi \left[\left(-\frac{1}{4} \right) - (-1) \right]$$

$$= \frac{3}{4}\pi$$

You can leave your answer in terms of π.

Example 21

Find the volume obtained when the shaded region is rotated through 360° about the x-axis.



Volume required = volume of cylinder, with $r = 3$ and $h = 4$, minus volume of revolution about x axis.

Work out a strategy for finding the required volume.

$y = \sqrt{2x+1}$

$$V = \pi \int y^2 \, dx = \pi \int_0^4 (2x + 1) \, dx$$

Don't forget to square y and put values of x for the bounds.

$$= \pi \left[x^2 + x \right]_0^4$$

$$= \pi \left[(4^2 + 4) - (0^2 + 0) \right]$$

Show the substitutions clearly.

$$= 20\pi$$

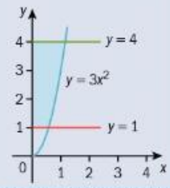
This gives us the volume when the area **below** the curve is rotated. The required volume is the volume of the cylinder found when rotating the line $y = 3$ about the x-axis with the volume of revolution removed. The volume of the cylinder is $\pi r^2 h$ where $r = 3$ and $h = 4$.

Required volume $= \pi \times 3^2 \times 4 - 20\pi = 16\pi$

Leave answer in terms of π unless told otherwise.

Example 22

The diagram shows the line $y = 1$, the line $y = 4$ and part of the curve $y = 3x^2$.
The shaded region is rotated through 360° about the y-axis.
Find the exact value of the volume of revolution obtained.



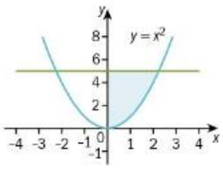
$y = 3x^2$, $x^2 = \dfrac{y}{3}$ — Rearrange to $x^2 = \ldots$

$$V = \int_a^b \pi x^2 \, dy = \pi \int_1^4 \frac{y}{3} \, dy$$ — Use the correct formula.

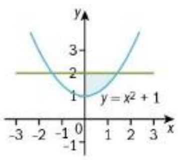
$$= \pi \left[\frac{y^2}{6} \right]_1^4$$

$$= \pi \left[\left(\frac{16}{6} \right) - \left(\frac{1}{6} \right) \right]$$

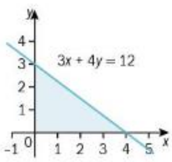
$$= \frac{15}{6}\pi = \frac{5\pi}{2}$$ — The question asked for the exact answer so leave the answer in terms of π.

3. 

Find the volume obtained when the shaded region is rotated through 360° about the y-axis.

4. 

Find the volume obtained when the shaded region is rotated through 360° about the y-axis.

5. 

Find the volume obtained when the shaded region is rotated through 360° about the x-axis.

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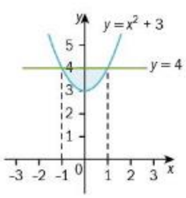
cor

6. Find the volume obtained when region bounded by the curve $y = x(3 - x)$ and the x-axis is rotated through 360° about the x-axis.

7. 

Find the volume obtained when the shaded region is rotated through 360° about the x-axis.

Summary:

Integrating x^n and ax^n

- $\int x^n \, dx = \frac{x^{n+1}}{n+1} + c$, provided $n \neq -1$.
- $\int ax^n \, dx = \frac{ax^{n+1}}{n+1} + c$, provided $n \neq -1$.

Integrating $\int (ax+b)^n \, dx$

- $\int (ax+b)^n \, dx = \frac{1}{a}\frac{(ax+b)^{n+1}}{(n+1)} + c$, provided $n \neq -1$.

The definite integral

- $\int_a^b f'(x)\,dx = [f(x)]_a^b = f(b) - f(a)$

Improper integrals

- When we have a definite integral with an upper limit of ∞ or a lower limit of $-\infty$ then we have an **improper** integral.
- When we have an integral where the function to be integrated is not defined at a point in the interval of integration then we have an **improper** integral.
- We define the improper integral $\int_a^{\infty} f(x)\,dx$ as $\lim_{b\to\infty} \int_a^b f(x)\,dx$, provided the limit exists.
- We define the improper integral $\int_{-\infty}^{b} f(x)\,dx$ as $\lim_{a\to-\infty} \int_a^b f(x)\,dx$, provided the limit exists.
- When $f(x)$ is defined for $0 \le x < b$, but $f(x)$ is not defined when $x = 0$, then the **improper integral** $\int_0^b f(x)\,dx$ is defined as $\lim_{a\to 0^+} \int_a^b f(x)\,dx$, provided the limit exists.